



University of Science and Technology in Zewail City (UST-ZC)
Communications and Information Engineering (CIE)

CIE 442 – Digital Signal Processing

Project 2: Digital Filter Design

Prepared By

Abdelhady Mohamed 202201172
Osama Mohamed 202201196

Instructor

Dr. Mahmoud Abdelaziz
Assistant Professor
Communications and Information Engineering Program

Contents

1	Problem Definition and Specifications	2
1.1	Problem Formulation	2
1.2	Constraints Identification	2
1.3	Filter Specifications	2
2	Filter Design and Analysis	3
2.1	Filters specifications	3
2.2	Filter Designs	3
2.2.1	IIR Butterworth Notch Filter	3
2.2.2	IIR Elliptic Notch Filter	4
2.2.3	FIR Equiripple Notch Filter	5
2.2.4	FIR Least-Squares Notch Filter	6
2.3	Decision Analysis	7
2.3.1	Criteria Weights	7
2.3.2	Decision Matrix	7
2.3.3	Final Selection	7
2.4	Implementation and Validation	7
3	Performance Evaluation	8
3.1	Frequency Domain Analysis	8
3.1.1	Frequency Spectrum Using Welch's Method	8
3.1.2	Spectrograms (STFT)	9
3.1.3	Effect of Window Types and Sizes	9
3.1.4	Removal of Undesired Components	10
3.2	Time Domain and Perceptual Evaluation	10

1 Problem Definition and Specifications

1.1 Problem Formulation

Electroencephalogram (EEG) signals represent the electrical activity of the brain and are dominated by low-amplitude, low-frequency components, mainly below 100 Hz. Because of their small amplitude, EEG recordings are highly susceptible to external interference.

A major source of contamination is power-line interference, which appears as a narrowband sinusoid at 60 Hz (or 50 Hz in some regions). This produces a strong spectral peak that overlaps with physiologically meaningful EEG bands and must be removed prior to further processing.

The objective of this project is to design and implement a digital filter that suppresses the 60 Hz interference while preserving the underlying EEG information. Since the interference is narrowband and neighbouring frequency content must be retained, a band-stop (notch) digital filter is the most appropriate choice.

1.2 Constraints Identification

The EEG signal is sampled at 256 Hz, which constrains the achievable frequency resolution. The filter order must also remain moderate to avoid unnecessary computational cost, particularly when processing long multi-channel EEG recordings.

Because EEG interpretation depends on waveform integrity, excessive amplitude or phase distortion is undesirable. Therefore, extremely high-order designs were avoided, and practical filter structures with stable behaviour and reasonable complexity were selected.

1.3 Filter Specifications

A narrow stopband centred at 60 Hz was specified, with approximate edges at 59–61 Hz, in order to target the power-line interference while preserving neighbouring EEG spectral content. A passband ripple limit of 0.5 dB was chosen to minimise distortion of physiological frequency bands, and a minimum stopband attenuation of 40 dB was specified to ensure effective suppression of the interference. Transition bandwidths were kept narrow while maintaining practical implementability for both IIR and FIR designs.

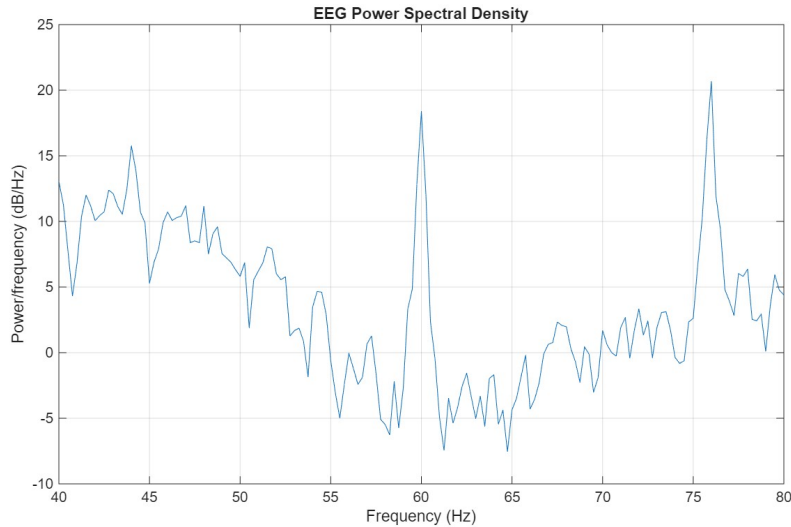


Figure 1: Original EEG Signal

2 Filter Design and Analysis

2.1 Filters specifications

Four performance criteria were used to evaluate and compare the designed filters. These criteria reflect both technical filtering performance and practical considerations in EEG signal processing.

Stopband Attenuation Stopband attenuation measures how strongly the filter suppresses unwanted frequency components inside the stopband. In this project, it determines how effectively the 60 Hz power-line interference is removed. Insufficient attenuation would leave residual noise in the EEG spectrum. Therefore, the goal is to maximise attenuation at 60 Hz, while avoiding unnecessary filter complexity.

Passband Distortion (Ripple) Passband distortion refers to amplitude variations within the desired EEG frequency band. Ripple alters the true spectral power distribution and may bias EEG interpretation. Since EEG analysis depends on accurate band power, passband ripple should be minimised in order to preserve the physiological information contained in the signal.

Phase Linearity Phase linearity indicates whether all frequency components experience equal delay. Non-linear phase behaviour distorts waveform shape by time-shifting different frequency components. Although phase linearity is desirable to preserve EEG morphology, the present work focuses primarily on frequency-domain analysis, so moderate phase distortion is acceptable provided filtering performance remains satisfactory.

Filter Order and Computational Efficiency Filter order determines computational complexity and memory requirements. Higher order filters require more processing time and may introduce numerical sensitivity. Since EEG recordings are typically long and multi-channel, it is advantageous to employ a filter that achieves the required performance with a reasonably low order.

2.2 Filter Designs

2.2.1 IIR Butterworth Notch Filter

A fourth-order IIR Butterworth band-stop filter was designed to suppress 60 Hz power-line interference. The filter parameters are summarized below:

- Filter type: IIR band-stop (notch)
- Design method: Butterworth
- Filter order: 4
- Sampling frequency: 256 Hz
- Notch center frequency: 60 Hz
- Stopband range: 59–61 Hz
- Phase response: Nonlinear

The frequency response of the filter as shown next:

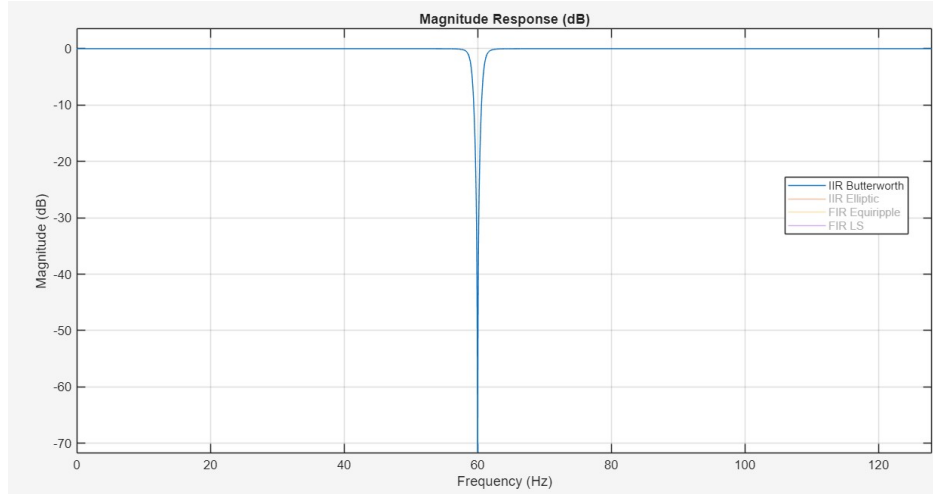


Figure 2: IIR Butterworth band-stop filter

2.2.2 IIR Elliptic Notch Filter

An IIR elliptic band-stop filter was designed to achieve sharp attenuation around 60 Hz with minimal filter order. The design parameters are as follows:

- Filter type: IIR band-stop (notch)
- Design method: Elliptic
- Filter order: 4
- Sampling frequency: 256 Hz
- Stopband range: 59.5–60.5 Hz
- Passband edges: 58 Hz and 62 Hz
- Passband ripple: 0.5 dB
- Stopband attenuation: 40 dB
- Phase response: Nonlinear

The frequency response of the filter as shown next:

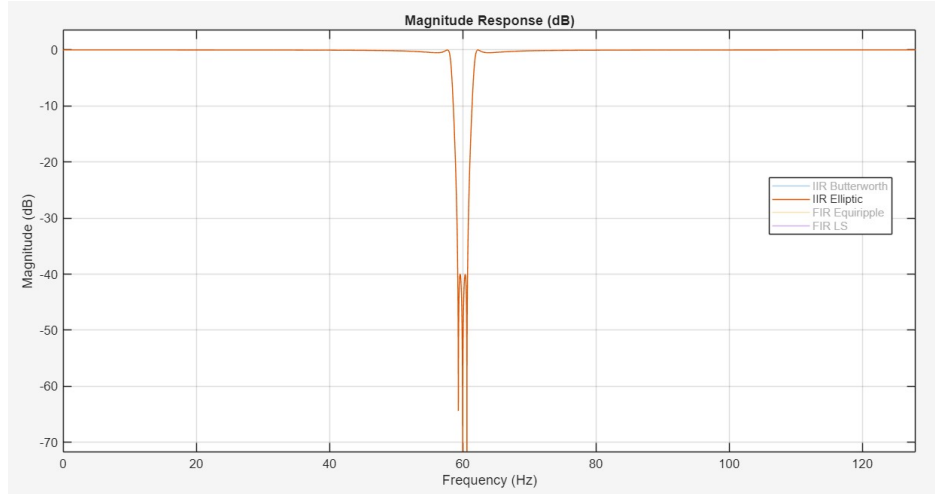


Figure 3: IIR Elliptic Notch Filter

2.2.3 FIR Equiripple Notch Filter

A linear-phase FIR equiripple band-stop filter was designed to suppress 60 Hz interference while preserving EEG waveform morphology. The filter parameters are listed below:

- Filter type: FIR band-stop (notch)
- Design method: Equiripple
- Filter order: 120
- Sampling frequency: 256 Hz
- Stopband range: 59–61 Hz
- Passband edges: 55 Hz and 65 Hz
- Passband ripple: 0.5 dB
- Stopband attenuation: 40 dB
- Phase response: Linear

The frequency response of the filter as shown next:

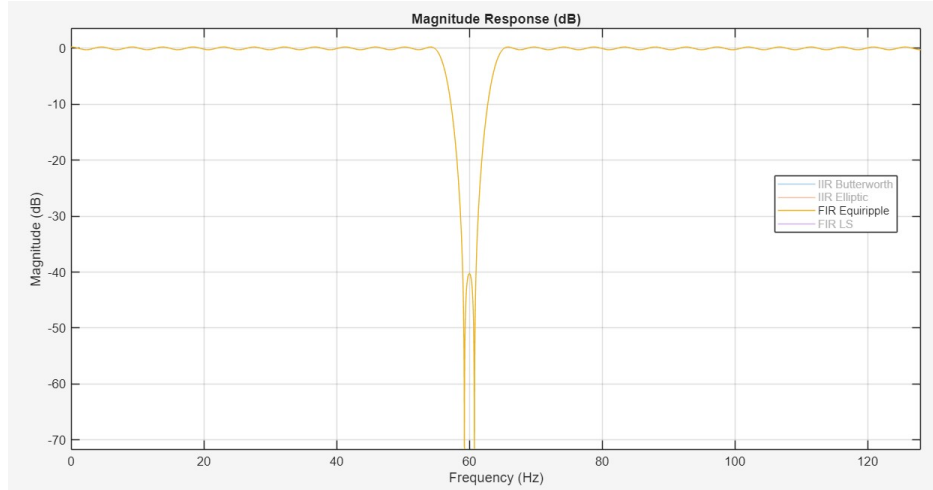


Figure 4: FIR Equiripple Notch Filter

2.2.4 FIR Least-Squares Notch Filter

An FIR least-squares band-stop filter was designed to minimize the mean-square error across frequency bands while maintaining linear phase. The design parameters are summarized below:

- Filter type: FIR band-stop (notch)
- Design method: Least-squares
- Filter order: 120
- Sampling frequency: 256 Hz
- Stopband range: 59–61 Hz
- Passband edges: 55 Hz and 65 Hz
- Phase response: Linear

The frequency response of the filter as shown next:

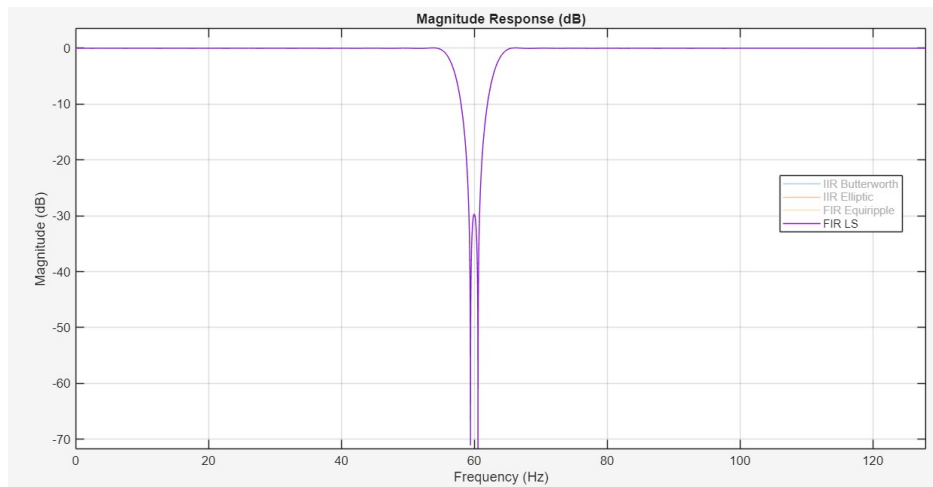


Figure 5: FIR Least-Squares Notch Filter

2.3 Decision Analysis

2.3.1 Criteria Weights

Each criterion was assigned a weighting factor according to its relative importance in achieving the project objectives. The weights sum to unity.

Criterion	Weight	Justification
Stopband Attenuation	0.35	Primary requirement is 60 Hz suppression
Passband Distortion	0.25	Preserve EEG spectral information
Phase Linearity	0.15	Desirable but not critical
Order & Efficiency	0.25	Practical and computationally efficient

Table 1: Weighting factors for decision criteria.

2.3.2 Decision Matrix

Each filter was scored from 1 (poor) to 5 (excellent) for each criterion based on its theoretical properties and observed behaviour. The final score for each filter was calculated as a weighted sum of the individual criterion scores:

$$S = w_{SB}A_{SB} + w_{PB}A_{PB} + w_{PH}A_{PH} + w_{EF}A_{EF}, \quad (1)$$

where A_{SB} , A_{PB} , A_{PH} , and A_{EF} denote the scores for stopband attenuation, passband distortion, phase linearity, and computational efficiency respectively, and w_{SB} , w_{PB} , w_{PH} , and w_{EF} are their corresponding weighting factors.

Filter	Stopband	Passband	Phase	Efficiency	Weighted Score
IIR Butterworth	5	4	2	5	4.30
IIR Elliptic	4	3	2	5	3.70
FIR Equiripple	3	1	5	2	2.80
FIR Least-Squares	4	2	5	2	2.90

Table 2: Decision matrix with weighted scoring (1 = poor, 5 = excellent).

2.3.3 Final Selection

Based on the weighted decision matrix, the IIR Butterworth notch filter achieved the highest overall utility score. It provides strong suppression of the 60 Hz interference with low passband distortion and excellent computational efficiency, without requiring a high filter order. Although FIR filters offer linear-phase behaviour, their significantly higher order results in increased processing cost with no substantial advantage for the present frequency-domain-based application.

Therefore, the IIR Butterworth filter was selected as the final design.

2.4 Implementation and Validation

The selected IIR Butterworth notch filter was applied to the EEG signal to suppress the 60 Hz power-line interference. The filter was implemented using zero-phase filtering to ensure accurate frequency-domain behavior. Validation was performed in the frequency domain using Welch’s power spectral density estimate. The results show a significant reduction of the spectral peak at 60 Hz after filtering, while the surrounding EEG frequency components remain largely unaffected. This confirms that the filter meets the design requirements and efficiently removes power-line interference. And the next image show it clearly:

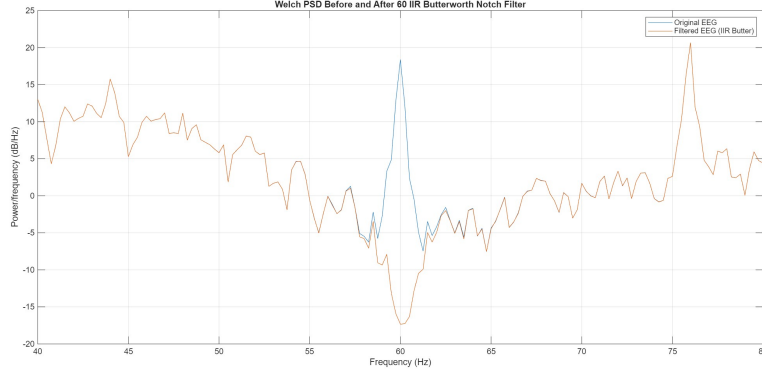


Figure 6: IIR Butterworth Notch Filter Effect

And the next Image shows the other filters effect compared to IIR Butterworth notch filter:

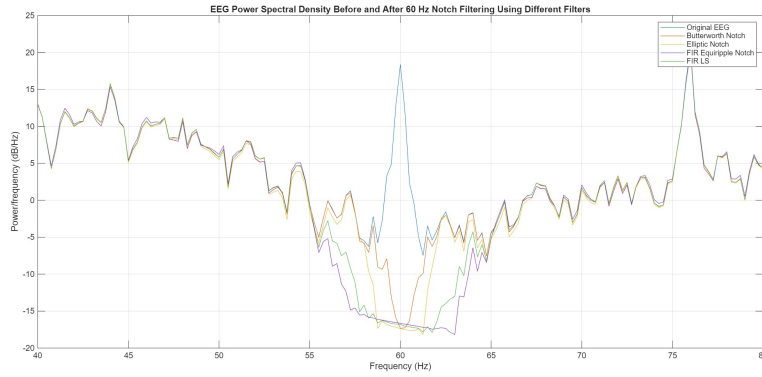


Figure 7: Different Notch Filters Effect

3 Performance Evaluation

3.1 Frequency Domain Analysis

3.1.1 Frequency Spectrum Using Welch's Method

Welch's power spectral density (PSD) estimation was used to analyze the frequency content of the EEG signal before and after filtering. This method provides a robust estimate of the signal spectrum by reducing variance through windowing and averaging. The PSD of the unfiltered EEG signal exhibits a prominent spectral peak at 60 Hz corresponding to power-line interference. After applying the selected IIR Butterworth notch filter, this peak is significantly attenuated, while the EEG frequency components of interest remain largely unaffected.

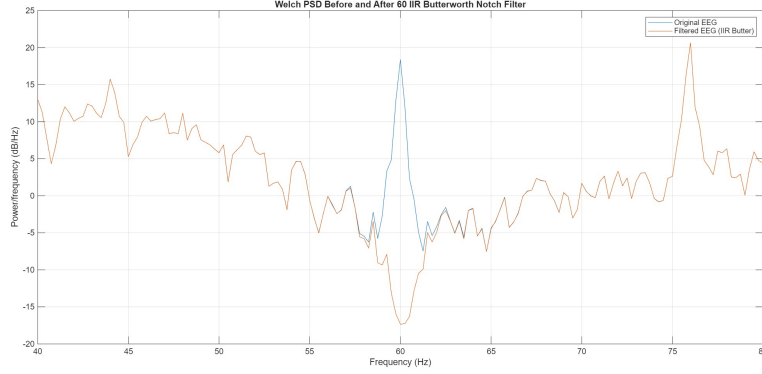


Figure 8: Frequency Spectrum Using Welch's Method

3.1.2 Spectrograms (STFT)

Short-time Fourier transform (STFT) spectrograms were used to analyze the time and frequency characteristics of the EEG signal. Spectrograms provide insight into how spectral components evolve over time. The spectrogram of the unfiltered EEG signal shows a persistent frequency component at 60 Hz. After filtering, this component is no longer present, indicating effective suppression of the power-line interference.

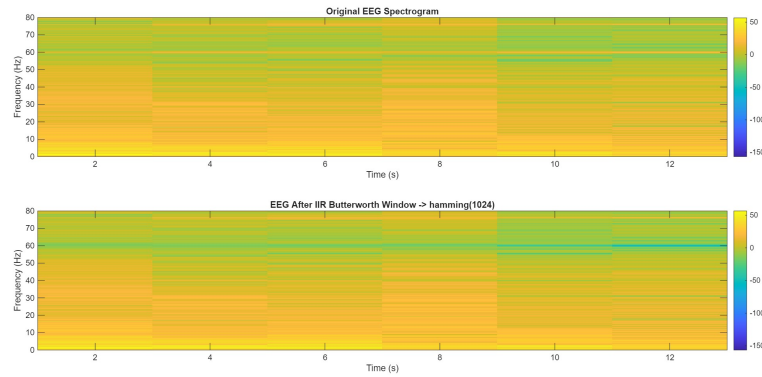


Figure 9: Spectrogram Before and After Filtration

3.1.3 Effect of Window Types and Sizes

The effect of different window types and window lengths on STFT analysis was investigated. Hamming and rectangular windows were used with varying window sizes to examine the trade-off between time and frequency resolution. Shorter windows provide improved time resolution but poorer frequency resolution, while longer windows enhance frequency resolution at the expense of temporal localization. Additionally, rectangular windows introduce increased spectral leakage compared to Hamming windows.

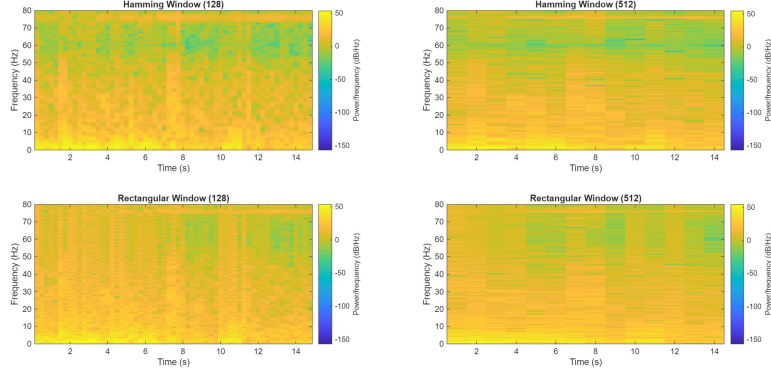


Figure 10: Window Effect on Spectrogram

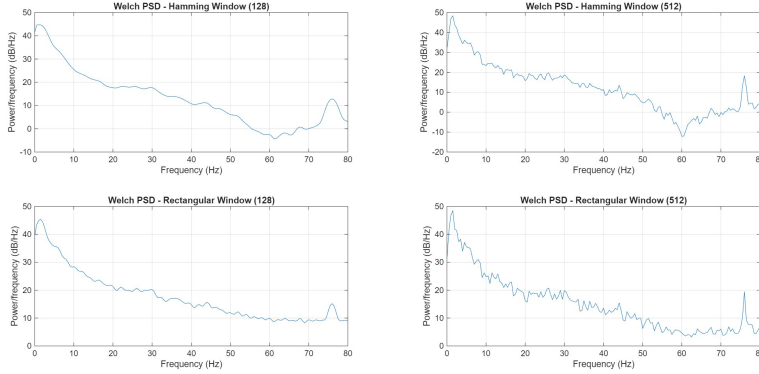


Figure 11: Window Effect on Welch

3.1.4 Removal of Undesired Components

The effectiveness of the designed filter was evaluated by examining the suppression of the undesired 60 Hz power-line interference. Comparison of the frequency-domain representations before and after filtering demonstrates that the undesired component is effectively removed. No significant attenuation is observed in neighboring EEG frequency bands, indicating that the filtering process preserves the signal of interest.

3.2 Time Domain and Perceptual Evaluation

Since the processed signal is an EEG recording, perceptual listening tests are not applicable. Therefore, evaluation was performed using time-domain signal analysis. The filtered EEG signal was visually inspected and compared with the original signal. The results show that the overall waveform structure and amplitude of the EEG signal are preserved after filtering. No noticeable distortions or artificial oscillations were introduced by the filtering process, indicating that the applied filter does not adversely affect the underlying neural activity.

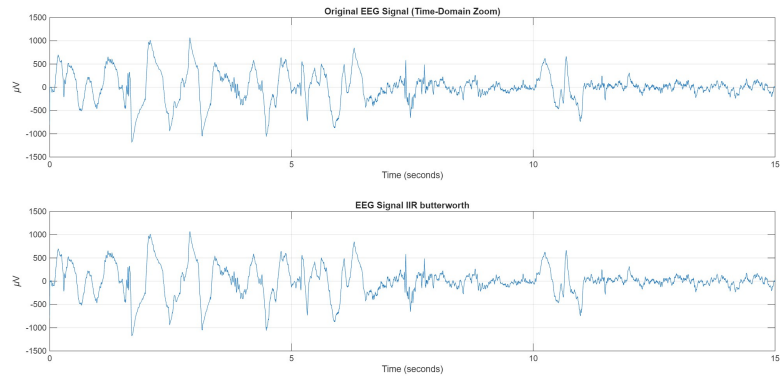


Figure 12: Time Domain Evaluation